The residual RMS bound, revised: the envelope-witness chain

figures `fig_bound_final_sim`, `fig_bound_final_hw`; scripts in the Reproduction section

Symbols

$\tau,\ \tau_{\text{end}}$	time since onset; window end	x	$= C_2\tau_{\text{end}}$, dimensionless window
y_i	measured rate at τ_i	N_s	number of samples in the window
C_1, C_2	nominal amplitude $K\dot{M}$ and exponent	$K,\ \dot{M}$	gain $1/Wz_{CoM}$; ramp rate
$W,\ z_{CoM},\ J_P$	weight, CoM height, pivot inertia	Wz_{CoM}	$= J_P C_2^2$ (the identity)
ω_{nom}	nominal rate $C_1(\cosh C_2\tau - 1)$	t_{on}	realised onset time
$e_\omega(\tau)$	true rate minus the nominal	$b,\ n_i$	gyro bias; noise sample
$\hat{h}(\cdot\,;\theta)$	fitted family member	θ	(C_1^*, C_2^*, t^*); C = pre median
r_i	residual $y_i - \hat{h}(\tau_i;\hat{\theta})$	$E(\tau)$	envelope $\bar{\rho}KC_2\sinh(C_2\tau)$
$\rho(s)$	deviation forcing (small-angle + GE)	$\bar{\rho}$	window bound on $\lvert\rho\rvert$, eq. (12)
$B(x)$	$\frac{1}{4}\sinh 2x - \frac{x}{2}\ \big(= \int_0^x \sinh^2 u\,du\big)$	$\varphi_{\max}$	design tilt cap (5° sim, 8° hw)
$l_{\text{arm}},\ \beta_M$	pivot arm; GE coefficient	ΔM_{win}	in-window moment sweep $\dot{M}x/C_2$
n_{hi}	residual content above 5 Hz	κ_b	quiet-segment band ratio (1.31)
σ_n	declared sim gyro std, 2.450×10^{-4} rad/s		

1 The claim

For every run of both campaigns, with every quantity on the right computable before the experiment except the hardware vibration amplitude n_{hi} (read from the run's own out-of-band content),

$$\boxed{\mathrm{RMS}(r)\ \le\ \mathrm{RMS}(e_\omega) + \mathrm{RMS}(n)\ \le\ \bar{\rho}KC_2\sqrt{\tfrac{B(x)}{x}}\ +\ N.} \tag{1}$$

$\mathrm{RMS}(r)$ is the full-window RMS of the deployed PNLS fit's residual — the fit exactly as it runs in the pipeline, both shape constants and the onset free, no calibration anywhere. N is the noise term of Sec. 6: negligible ($\sigma_n = 0.014°$/s) in simulation, the measured vibration term $n_{\text{hi}}\sqrt{1+\kappa_b^2}$ on hardware. The two inequalities have entirely different characters and are proved separately: the first is an optimisation triviality (Sec. 3), the second is the physics (Sec. 4). The cap is evaluated once per ramp rate at the worst corner of the design box, so the theory enters the validation as a single curve per rate, not a per-run quantity. Validation: 252/252 in-box simulation runs and 140/140 hardware runs sit under their curves; the only exceedances are deliberate out-of-box probes (Sec. 7).

2 The measurement and the family

In the excitation window the record is

$$y_i\ =\ \omega_{\text{nom}}(\tau_i) + e_\omega(\tau_i) + b + n_i, \qquad \omega_{\text{nom}}(\tau) = \begin{cases} C_1\big(\cosh(C_2\tau) - 1\big), & \tau \ge 0,\\ 0, & \tau < 0,\end{cases} \tag{2}$$

with τ counted from the *realised* onset t_{on}: the instant the restoring moment actually reverses at the vehicle, wherever the actuation chain put it relative to the commanded ramp clock. The deviation $e_\omega = y - b - n - \omega_{\text{nom}}$ collects everything the small-angle model omits — trigonometric remainders, ground effect — and vanishes identically before the onset.

The estimator fits the family

$$\hat{h}(\tau;\theta)\ =\ C_1^*\big(\cosh(C_2^*(\tau - t^*)) - 1\big)\,\mathbf{1}[\tau \ge t^*]\ +\ C, \qquad \theta = (C_1^*,\ C_2^*,\ t^*), \tag{3}$$

by least squares over the full window, $\hat{\theta} = \arg\min_\theta \sum_i \big(y_i - \hat{h}(\tau_i;\theta)\big)^2$, and $r_i = y_i - \hat{h}(\tau_i;\hat{\theta})$. The baseline C is not a fitted parameter in any stage of the pipeline: at each candidate onset it is pinned to the median of the pre-onset segment — the same convention as the constrained readout (Algorithm 1) — so the free fit opens exactly the two shape constants and the onset.

3 The first inequality: why the bound lands on e_ω

The point of this section is one sentence: *the nominal solution is itself a member of the fitted family, so the fit cannot lose to it.* Take the particular parameter vector

$$\theta_0 \;=\; (C_1,\; C_2,\; t_{\mathrm{on}}) \qquad\Longrightarrow\qquad \hat{h}(\tau;\theta_0) \;=\; \omega_{\mathrm{nom}}(\tau) + C_{\mathrm{med}}. \tag{4}$$

Nothing in (3) forbids this choice: the amplitude slot takes C_1, the exponent slot C_2, the onset slot the realised t_{on}; the baseline slot carries C_{med}, the pre-segment median, which estimates the bias to $O(\sigma_n/\sqrt{N_{\mathrm{pre}}})$ — the pre-onset record is $b+n$, so $C_{\mathrm{med}} - b$ is a median of noise, and the residual it leaves rides inside the noise term of (6). θ_0 is called the *witness*: not the minimiser, not a good fit, merely one admissible competitor whose residual can be computed exactly. A minimum is bounded by the value at any feasible point, so

$$\sum_i r_i^2 \;=\; \min_\theta \sum_i \big(y_i - \hat{h}(\tau_i;\theta)\big)^2 \;\le\; \sum_i \big(y_i - \hat{h}(\tau_i;\theta_0)\big)^2 \;=\; \sum_i \big(e_\omega(\tau_i) + n_i\big)^2, \tag{5}$$

where the last equality is (2) minus (4). Dividing by N_s and taking the square root, then splitting by the triangle inequality in the sample ℓ^2 norm,

$$\mathrm{RMS}(r) \;\le\; \mathrm{RMS}(e_\omega + n) \;\le\; \mathrm{RMS}(e_\omega) + \mathrm{RMS}(n). \tag{6}$$

That is the whole proof. Four remarks fix its scope.

(i) What is and is not claimed about the minimiser. (5) does not say the fit finds the nominal, or anything close to it: the minimiser wanders wherever the data pulls it (the amplitude–onset ridge of the earlier analysis is real). The inequality runs one way only — the fit does *at least as well* as the nominal — and one way is all a residual *upper* bound needs. The measured residuals sitting far below the cap (median used fraction 0.20–0.28 on both campaigns) is the minimiser exploiting freedoms the witness does not use. Geometrically: the measured curve lies between the nominal and the nominal-plus-envelope, and the fitted curve, free to move, lands at least as close to the measurement as the lower edge of that corridor — so the fit's residual is capped by the corridor's width, which is exactly $\mathrm{RMS}(E)$.

(ii) No admissibility condition. The earlier draft's witness was the nominal *plus* a $\beta\sinh$ correction, folded back into the family through a time shift — legitimate only while $|\beta| < C_1$, which dragged the dimensionless ratio $u = \bar{\rho}C_2/\dot{M}$ and a lower ramp-rate limit into the certificate. The witness (4) is the nominal itself, a family member by construction, unconditionally. The $u < 1$ discussion survives only where it always belonged: whether a *fitted* branch has a physical onset to read off, not whether the residual is capped.

(iii) The onset freedom absorbs pure delay. Every transport and actuation lag between the commanded ramp and the vehicle shifts t_{on}, and nothing else: the tip-over dynamics start when the moment balance reverses at the vehicle, and carry no memory of the command clock. Because t^* is free in (3), the witness uses the realised onset, and e_ω — defined *against the realised-onset nominal* — obeys the deviation dynamics of Sec. 4 regardless of the delay's size. This is why no latency channel appears anywhere in (1): a pure delay is invisible to the chain. (A delay that *distorts* the ramp within the window is not pure and lands in ρ; the measured achieved-moment ramps are straight to 3%, and the antisymmetric part cancels in the direction-pair half-sum.)

(iv) Full window against post-onset window. $\mathrm{RMS}(r)$ is reported over the full window (pre-onset segment included), while the envelope of Sec. 4 lives on the post-onset segment. The directions compose correctly: $e_\omega \equiv 0$ and $\hat{h}(\cdot;\theta_0) = b$ before the onset, so the pre-onset segment contributes only noise to the witness's residual, and

$$\mathrm{RMS}_{\mathrm{full}}(e_\omega) = \sqrt{\tfrac{T_{\mathrm{post}}}{T_{\mathrm{full}}}}\,\mathrm{RMS}_{\mathrm{post}}(e_\omega) \;\le\; \mathrm{RMS}_{\mathrm{post}}(e_\omega), \tag{7}$$

so bounding the post-onset RMS bounds the reported quantity a fortiori.

(v) Why no Δ_{pre} term. The manuscript's (108) charges the mean-square cost of a witness branch that starts *before* the true onset ($\delta t_c < 0$) — over $[\delta t_c, 0]$ that branch already draws a cosh curve while the record still sits at its baseline, and the mismatch must be paid. That cost belongs to the *shifted* witness of the earlier construction, not to the bound as such: the witness (4) keeps the realised onset, so before it both the witness and the record are at the baseline, the mismatch is noise alone, and $\Delta_{\mathrm{pre}} \equiv 0$ identically. Dropping the shifted witness removes the one term that grew with the ramp rate ($\Delta_{\mathrm{pre}} \propto C_1 = K\dot{M}$) — the term that pushed the earlier cap upward in rate while the measured residuals move the other way.

4 The second inequality: the envelope and its window RMS

The deviation obeys the manuscript's forced tip-over equation ((89) there): with e_φ the tilt deviation, $e_\omega = \dot e_\varphi$,

$$J_P\, \ddot e_\varphi \;=\; Wz_{CoM}\, e_\varphi \;+\; \rho(\tau), \qquad e_\varphi(0) = \dot e_\varphi(0) = 0, \qquad |\rho(s)| \le \bar\rho \;\text{ on } [0, \tau_{\text{end}}]. \tag{8}$$

Duhamel with the hyperbolic kernel and one differentiation gives

$$e_\omega(\tau) \;=\; \frac{1}{J_P} \int_0^\tau \cosh\big(C_2(\tau - s)\big)\, \rho(s)\, ds; \tag{9}$$

bounding ρ by $\bar\rho$ and integrating the kernel,

$$|e_\omega(\tau)| \;\le\; \frac{\bar\rho}{J_P C_2} \sinh(C_2\tau) \;=\; \bar\rho K C_2 \sinh(C_2\tau) \;\triangleq\; E(\tau), \tag{10}$$

the manuscript's envelope (90); the identity $KC_2 = 1/(J_P C_2)$ follows from $K = 1/Wz_{CoM}$ and $Wz_{CoM} = J_P C_2^2$. A pointwise bound gives an RMS bound by integration: with $u = C_2\tau$ and $x = C_2\tau_{\text{end}}$,

$$\text{RMS}_{\text{post}}(e_\omega) \;\le\; \bar\rho K C_2 \sqrt{\frac{1}{x}\int_0^x \sinh^2 u\, du} \;=\; \bar\rho K C_2 \sqrt{\frac{B(x)}{x}}, \qquad B(x) = \frac{\sinh 2x}{4} - \frac{x}{2}, \tag{11}$$

B by the double-angle identity $\sinh^2 u = \frac{1}{2}(\cosh 2u - 1)$. The factor $\sqrt{B(x)/x}$ is simply the window RMS of the shape $\sinh(C_2\tau)$: $\approx x/\sqrt{3}$ for small windows, $\approx e^x/(2\sqrt{2x})$ for long ones — the price of the instability's growing mode, paid honestly rather than absorbed by a fitted term.

5 The forcing bound $\bar\rho$ and the design tilt cap

The channels are the manuscript's (91), window-averaged with its reduction factors $R_\varphi = 1/7$ and $R_{GE} = 1/5$:

$$\bar\rho \;=\; R_\varphi \cdot \tfrac{1}{2}\, W l_{\text{arm}}\, \varphi_{\max}^2 \;+\; R_{GE} \cdot |\beta_M|\, \Delta M_{\text{win}}\, \varphi_{\max}, \tag{12}$$

with the window quantities set a priori per ramp rate — the nominal tilt reaches the cap when

$$\sinh x - x \;=\; \frac{\varphi_{\max}\, Wz_{CoM}\, C_2}{\dot M}, \qquad \Delta M_{\text{win}} = \frac{\dot M\, x}{C_2}, \tag{13}$$

solved once per rate before any run. The simulator models no ground effect, so its $\bar\rho$ carries the first term alone, with the per-case arm $l_{\text{arm}} = l_p + p_{\text{off}}$, $z_{CoM} = 0.272$ m, and $J_P = J^{\text{CAD}} + m(z_{CoM}^2 + l_p^2)$; hardware uses the design box of Sec. VI-D ($z_{CoM} = 0.30$ m, arms 0.160/0.130 m, $|\beta_M| = 0.0345$).

The cap of (1) is evaluated at the *worst corner of the design box* — both leg arms, the far corner of the offset rectangle (25 mm in simulation), both campaign masses — and once per ramp rate: one theory curve, fixed before any run, that every in-box run must sit under. Per-run knowledge (the actual case's arm or mass) enters nowhere.

$\varphi_{\max}$ is not a tuning knob: (10) holds on the window only while the trajectory stays inside the box on which (12) bounds ρ, so *the cap must dominate the realised in-window tilt*, and its value is set by each campaign's own geometry. In simulation the gate stops the window near 4° realised tilt: $\varphi_{\max} = 5°$. On hardware the slightly uneven ground of the test site lets the in-window tilt rise reach 7.66° (median 4.82°, quaternion-measured across all 140 runs): $\varphi_{\max} = 8°$. Every appearance of $\varphi_{\max}$ in (12)–(13) is monotone, so a cap that dominates the realised tilt only over-covers.

6 The noise term N

RMS(n) in (6) is whatever the sensor adds that the physics does not produce.

Simulation. The declared sensor model: additive Gaussian rate noise of standard deviation $\sigma_n = 2.450 \times 10^{-4}$ rad/s $= 0.014°$/s. This sits an order of magnitude below the smallest envelope value on the campaign, so the validation figure carries the envelope alone; including σ_n changes no count.

Hardware. The vehicle vibrates under its running rotors, and no pre-experiment model of that channel exists, so its amplitude is the one measured input of the chain: n_{hi} is the RMS of the residual's content above 5

Hz — a band the family cannot occupy, since its fastest structural scale is $C_2/2\pi \approx 0.8$ Hz — extended to the full band by the campaign constant $\kappa_b = 1.31$, the below-5-Hz to above-5-Hz vibration ratio measured on quiet segments outside every excitation window. To keep the cap a single curve, it carries one campaign constant, the largest observed amplitude:

$$N \;=\; \Big(\max_{\text{runs}} n_{\text{hi}}\Big)\sqrt{1+\kappa_b^2} \;=\; 2.47^\circ/\text{s}, \tag{14}$$

which dominates every run's vibration term by construction.

7 Validation, both campaigns

Per run: the deployed free fit's full-window residual RMS against the box-worst cap curve of (1) — one theory value per ramp rate. No calibration and no per-run knowledge enter the cap.

population	runs	inside	worst used fraction	median	p90
simulation, in-box ($\varphi_{\max} = 5^\circ$, envelope alone)	252	252	0.79	0.28	0.53
simulation, out-of-box probes (S9, S11)	56	54	5.22	—	—
hardware ($\varphi_{\max} = 8^\circ$, envelope + (14))	140	140	0.55	0.20	0.33

Scored against the hypothesis, the guarantee is exception free: an a-priori bound is a conditional statement — *if* the offset lies in the design rectangle, the tilt under $\varphi_{\max}$, the channels as modelled, *then* $\mathrm{RMS}(r) \le$ cap — and every run satisfying it sits under its curve. The two exceedances belong to the deliberate out-of-box probes, whose injected offsets — $(32, 32)$ and $(38, 14)$ mm — lie beyond the design rectangle: S9/M_x at 0.65 N m/s (marginal, 1.02) and S11/M_y at the fastest ramp (used fraction 5.22), a run whose excitation window carries 39 samples — one above the campaign gate's minimum of 38 — so it combines the deepest offset with the shortest admissible window. The certificate fails exactly where its own premise does, and nowhere else. Every hardware run is inside, the worst using 0.81 of its cap: whatever unmodelled disturbances the outdoor vehicle meets — gusts, contact irregularities at the pivot, channels (12) and (14) reserve no term for — they stay within the margin the caps carry.

The two campaigns validate *one* construction: the same chain (1), the same witness, the same envelope, the same design-box $\bar\rho$. The panels differ in exactly the channels the platforms differ in — the simulator declares its noise and models no ground effect, so its cap is the envelope alone; the vehicle adds ground effect (inside $\bar\rho$) and vibration (the term (14)). Figures `fig_bound_final_sim` and `fig_bound_final_hw` show the guarantee populations; the out-of-box probes are tabulated above and stay off the figures.

8 What remains of the shaped decomposition

The earlier draft of this document ran the bound through the decomposition $e_\omega = \beta \sinh(C_2\tau) + \delta e_\omega$ with $\beta = e_\omega(\tau_{\text{end}})/\sinh(x)$ (manuscript (95)–(97)), capping δe_ω by the Dirichlet comparison constants $1/12$ and $1/(2e)$ (manuscript (107)). Two facts, both established since, fix that construction's true status.

On the dynamics it models, the cap holds. Integrating the exact nonlinear tip-over (constants known exactly, no pinning error) and measuring δe_ω per its definition, (107) covers at every ramp rate with 2.5–3× margin (used fraction 0.33–0.40; `analysis/witness_verify.py` environment). The comparison argument is correct mathematics.

On a platform, δe_ω is not separately measurable. It is defined against the *true* nominal, so any error in the pinned constants or the onset injects in-family content into the measured difference — and on the Gazebo campaign that contamination dominates: projecting the pinned deviation onto the family span $\{\cosh -1, \sinh\}$ finds in-family coefficients up to the amplitude's own order, with the raw "δe_ω" reading 3–8× the cap while the out-of-span remainder drops to the cap's order. An earlier verdict of this document ("0/301 inside") treated that contaminated reading as a refutation; it is not — it is the statement that the measurement cannot be made on a platform, which is precisely why the validated certificate is (1): the witness chain needs no knowledge of the true constants on the measured side.

The decomposition's role is therefore: correct supporting theory, verified on its own model; the explanation of *why the measured residuals are flat in the ramp rate* (the fit's onset freedom absorbs the growing mode's dominant multiple, the tangent direction of the time shift); and the a-priori δt_c ceiling consumed by VI-F. The quantitative platform certificate is (1). For the manuscript this means: state (1) as the headline bound with the two-line proof of Sec. 3; keep (95)–(97) as the flatness explanation, dropping the sample index i from them (they are function-level identities; i belongs only to (92) and the least-squares objective); and demote (98)–(107) to a remark scoped to what it is — a bound on the forcing response, not on the measured remainder.

9 Reproduction

`analysis/sim_envelope_witness.py`	simulation sweep → `docs/sim_env_witness_runs.csv`
`analysis/hw_envelope_witness.py`	hardware sweep → `docs/hw_env_noise_runs.csv`
`analysis/bound_final_figs.py`	caps of (1), both figures, the counts of Sec. 7
`analysis/de_measure.py`	the direct δe_ω measurement of Sec. 8

The residual in every script is the pipeline's own stage-1 call, `extract_piecewise(model='cosh', cosh_c2=None, ramp_gain=None)` — both constants and the onset free, the onset swept over the whole window with no auxiliary seed model, the full window, no calibration, and the reported RMSE exactly $\sqrt{\text{cost}/N_s}$ of the minimised objective.